

## Smart grids: A smart turn for India

A smart grid is typically a collection of smart devices and applications applied to the electricity grid to provide real-time information about the performance of the grid. This information allows the service provider (utility) to make smarter and faster decisions in order to improve productivity and power management. When extended to the customers, it enables 'demand-side' management—and a green home. That is, it equips customers with the information required to understand and control their power consumption, thereby helping to reduce the footprint of each household!

Last year, North Delhi Power Limited (NDPL) engaged GE to implement a smart grid focused on power outage management. The smart grid solution would help to maximise the system's uptime, while also helping NDPL to quickly restore power to homes and businesses in the event of an outage. This delivery optimisation would improve reliability, and also reduce losses, thanks to better visibility and management.

While the current project is aimed more at power outages, demand-side management is also planned as part of the Restructured Accelerated Power Development and Reforms Programme (RAPDRP). If implemented over time, it would have a blanket effect, enabling several buildings to become green by simple power management!

Another interesting technology was announced by GE last year, in the context of the smart grid—the zero net-energy home project, through which GE will show that it is possible to build houses that produce as much energy as they consume, at just 10 per cent additional cost. The model will use GE's technologies and smart appliances for generating, storing and managing energy. Connected to a smart grid, the house will be in a position to not just buy power from the grid but also sell excess solar energy!

undeniable proof of the potential of technology in green buildings.

### How green is my valley?

"Green" is an often-used term these days. Green is, broadly, anything that is environment-friendly. It is used to popularise energy-saving practices in daily life without sacrificing the quality of life. There are two sides to this statement. One, the consumption of less energy through better efficiency, thereby reducing the production levels, and the other, the production of energy in

an environment-friendly manner. Both these aspects put together will almost guarantee smaller carbon footprint," says Ramprasad Ananthaswamy, director-power management products, Texas Instruments India.

There are several ways in which a building can be made green—right from the design, the materials used and methods of construction to the technologies (design tools, sensors, smart meters, temperature controls and timers, lighting systems, energy management solutions, etc) built into

it. On another plane, the occupants of an existing building can also add to the green hue by adopting energy-efficient gadgets and appliances, harvesting rainwater and so on.

This concept of 'green building' as such is quite nascent—but in every way possible—in India. There are some splendid examples including full-fledged complexes such as the Confederation of Indian Industries' Green Business Centre in Hyderabad and the recently redone Thyagaraj Stadium in New Delhi.

The expertise, the technologies and the products required for sustainable 'green' buildings are aplenty today. "First, we had problems with the availability of these products; now these are easily available. Cost is separate from value. The cost of these equipment is, like in any other trade, a factor of demand and consumption. We have seen the prices of these equipment fall over the past few years. These are currently available at prices that are within the reach of the so-called upper middle-class families. The not-so-highly-technological items like fluorescent lamps and gas geysers are priced such that even a lower middle-class person can afford. In my understanding, the fall rate is adequate as of now, and it can only be accelerated with a parallel acceleration in usage," says Nilesh Gandhi, architect and sustainable building design specialist.

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